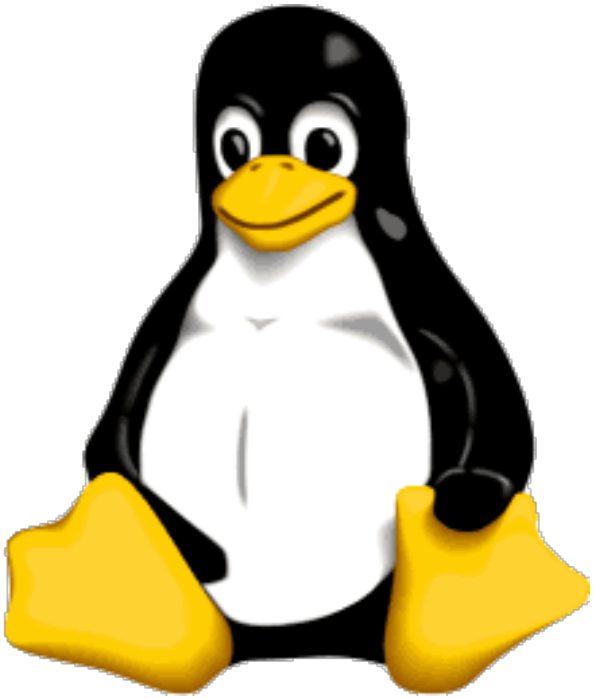
IOT Without Things: Building virtual 6LoWPAN Mesh Networks

netdev 2.1

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6LoWPAN L2 Subsystems

- IEEE 802.15.4 (This talk only!)
  - Mesh
  - (Try) to use net_device :-)
- BTLE 4.x
  - Point to Point
  - Use hci_dev :-(
- Bluetooth 5.x (new)
  - More Mesh
  - Answer to IEEE 802.15.4



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Situation

IEEE 802.15.4 isn't widely used (mostly by research groups)

Actual Spec: 802.15.4-2015

IETF is specifies a lot of RFC for 6LoWPAN in combination with this Link Layer



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Community Work

We working together with Communities!

(Mostly RIOT-OS (LGPL))

Others (not so much contact on ML):

- Zephyr (Other OS by collaborative LF)
- Contiki
- Mbed (Some parts closed)
- OpenThread (Open impl. of Thread)



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Community Problems

My expierence with RIOT-OS

- Users detect issues
- (Me) Gets always cc’ed
- Networking experts tries to follow
- Hard to reproduce issue
- Without reproduce → It takes longer



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Development Problems

- Mainly Routing Protocols
  - IEEE 802.15.4 → No L2 routing
  - RPL: Low-Power and Lossy Networks
  - Manipulate PHY Layer in real world
- Fuzzing Stack for kernel crashes
- IOT Application Development

Or simple try it out without buying hardware!



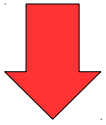
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Related Work: Other Setups

Mostly virtual UDP based

Handle IEEE 802.15.4 frames as UDP payload and feed application stack



Complex and difficult to connect together



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Solution: PHY Layer in Kernel

Emulating ONE PHY Layer (Kernel Driver: FakeLB)

- Checks Channel settings
- Runtime Manipulation
- Connect User Space Stack (IOT OS)
- One PHY to catch all traffic (Sniffer)



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User Space Stacks

IOT OS “native” platforms

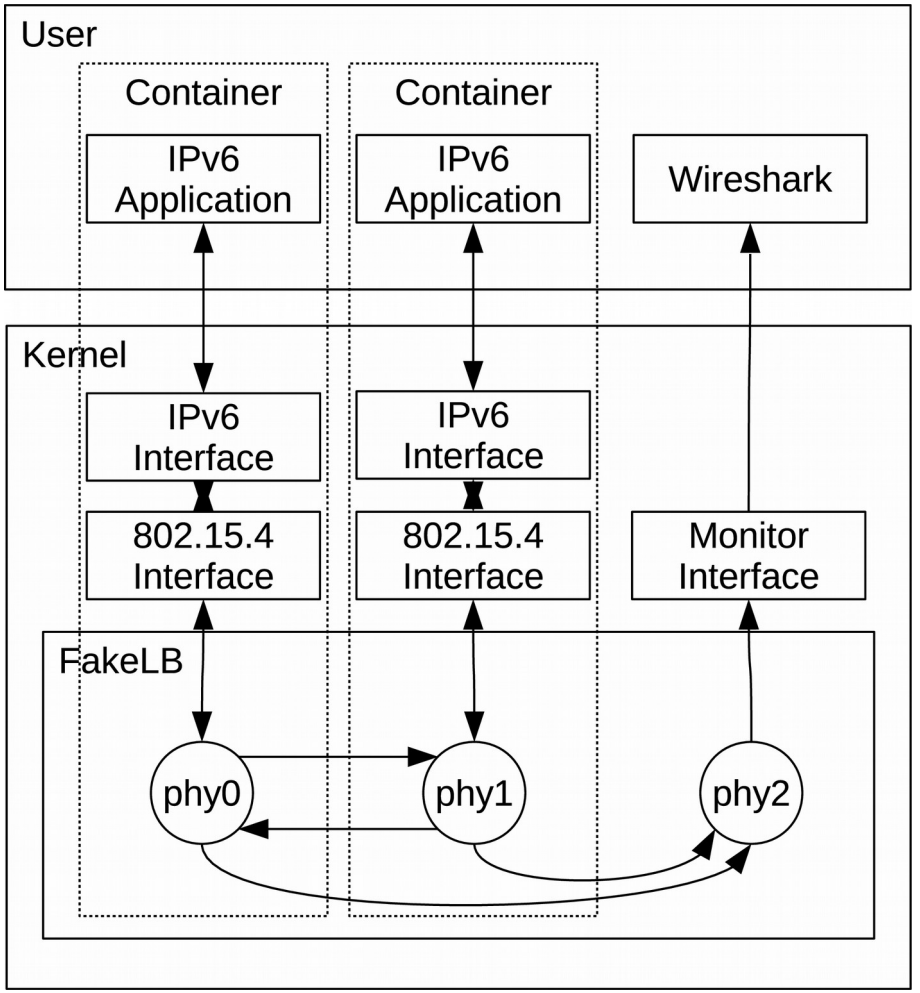
- Executable ELF-Binary
- PHY Configuration (nl802154)
- For Frames AF_PACKET RAW
- Monitor Interfaces (No MAC)
- Shell access to IOT-OS



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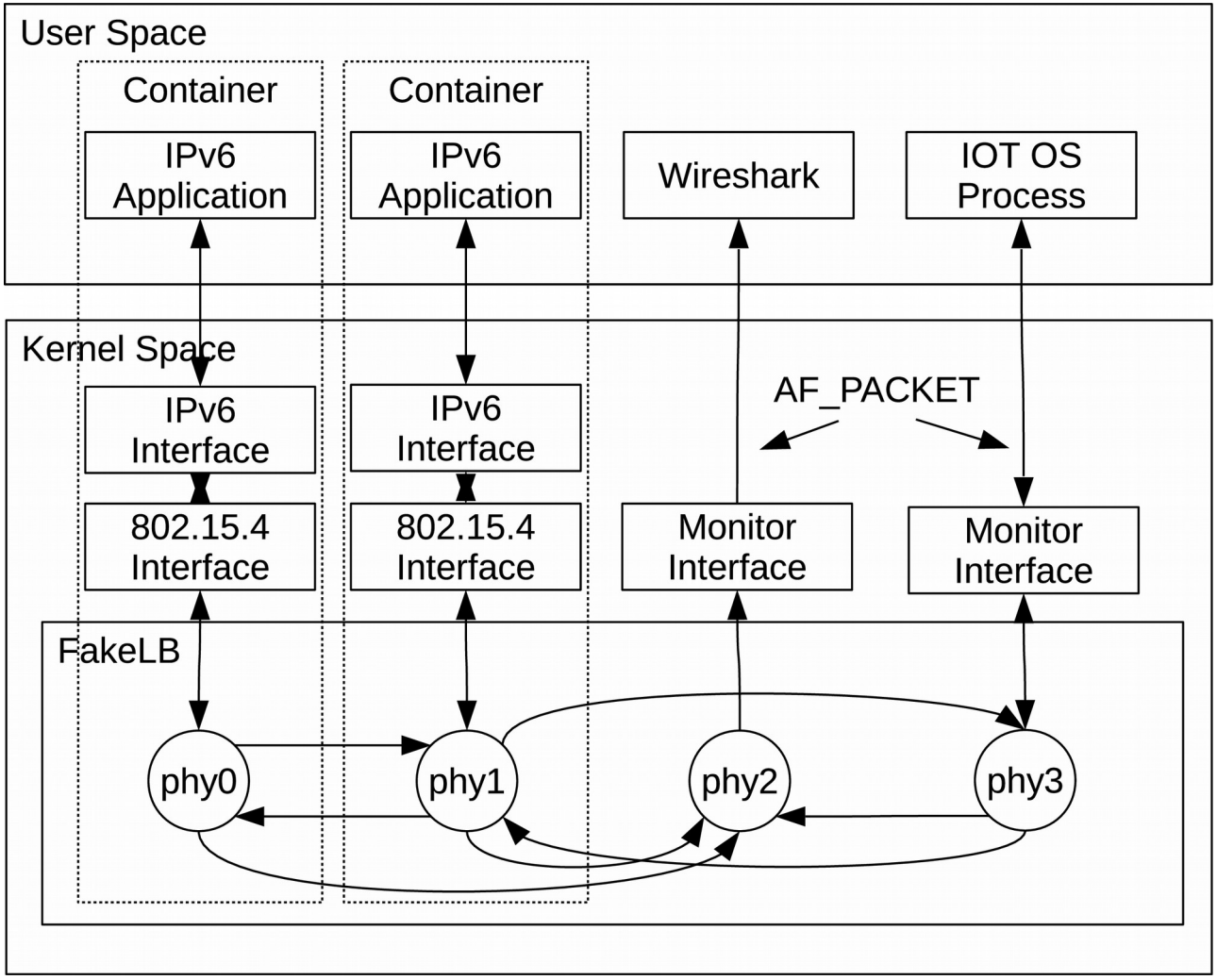
How it works: Linux only



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How it works: With IOT OS



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How it works

All PHYs are still IEEE 802.15.4 PHY Device Class

Means: IOT OS can run on real hardware from user space
(we don’t recommend it)

Supported:
RIOT OS/OpenThread



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How it works

How to connect PHYs?

- FakeLB UAPI and PHY Index
  - $echo “add 1 2” > .../edges
  - $echo “del 1 2” > .../edges
  - $echo “lqi 1 2 0xbb” > ../edges
- Scripts to reproduce Setup



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Future Work

Only few of my ideas

- Per Frame PHY Information
- IOT OS Driver (nl802154)
- QEMU IOT OS connection?
- Automated Testing?



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Demo Time



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